

Financial Risk Engine

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2. Problem and Approach

The Challenge:

Modern markets react to unstructured events faster than price tickers update. Traditional portfolio management often relies on lagging indicators, exposing portfolios to unseen risks.

Our Approach:

Build an event-driven risk-intelligence prototype that:

1. Ingests financial news in real-time.
2. Quantifies sentiment and calculates a heuristic risk score.
3. Dynamically rebalances a stock index based on real-time news constraints.
4. Stress tests portfolios across both trading and banking books.

3. System Architecture

Data Sources: Google News RSS + Synthetic Social Feeds

Ingestion & Deduplication: SQLite backend with hash-based entity resolution

Core Engine:

- **Risk NLP:** TF-IDF + Logistic Regression fallback generating quantifiable signals.
- **Module A:** Tactical Index Rebalancer (Constraint-based tilting).
- **Module B:** Stress Testing Engine (Macro sensitivities and Credit impacts).

Presentation: Streamlit + Plotly Dashboard

4. Data Sources and NLP Design

- **Engineering Trade-off (FinBERT vs. TF-IDF):** We considered FinBERT for domain-aware sentiment. However, due to large model sizes, memory constraints, and the need for reliable offline hackathon execution, we deliberately selected a lightweight **TF-IDF + Logistic Regression** model.
- **Social Media Limitations:** The fallback model is trained on formal financial news (`all-data.csv`). Social media text (slang/hashtags) is ingested but processed through this news-trained model; dedicated social preprocessing is a future iteration.
- **Source Provenance:** Strict delineation between live feeds (RSS/Social) and Offline Replay data, with accurate taxonomy deduplication preventing duplicate rebalancing.

5. Demonstration Results

Working Validations:

- Offline Crisis Replay successfully mocks historical shocks, generating immediate target allocation updates.
- Auto-triggered stress charts dynamically display when severe impact signals are detected (Impact > 7).
- Both primary Google RSS and secondary Synthetic Social data are demonstrably processed and uniquely tracked.

Note: Outcomes rely on prototype algorithms and do not represent verified financial alpha.

6. Domain Impact

- **Event-Driven Resilience:** Directly integrates unstructured NLP to mitigate risk faster than waiting for standard moving average crossovers.
- **Holistic View:** The dual inclusion of Trading Book (Equities) and Banking Book (Synthetic Credit Loans) demonstrates cross-asset maturity.
- **Auditability:** Transparent attribution trails (e.g. knowing whether a signal came from an LLM or Keyword Heuristic) are critical for regulatory compliance in modern fintech.

7. Limitations and Next Steps

- **Current Limitations:**

- Heuristic score is not a calibrated statistical probability.
- Stress approximations use first-order linear models and lack convexity.
- Market liquidity impacts are assumed constant.

- **Next Steps:**

- Introduce calibrated confidence scores via historical backtesting.
- Implement full non-linear pricing models for complex credit derivatives.
- Expand ingestion beyond top 10 tickers to universe-wide scans.